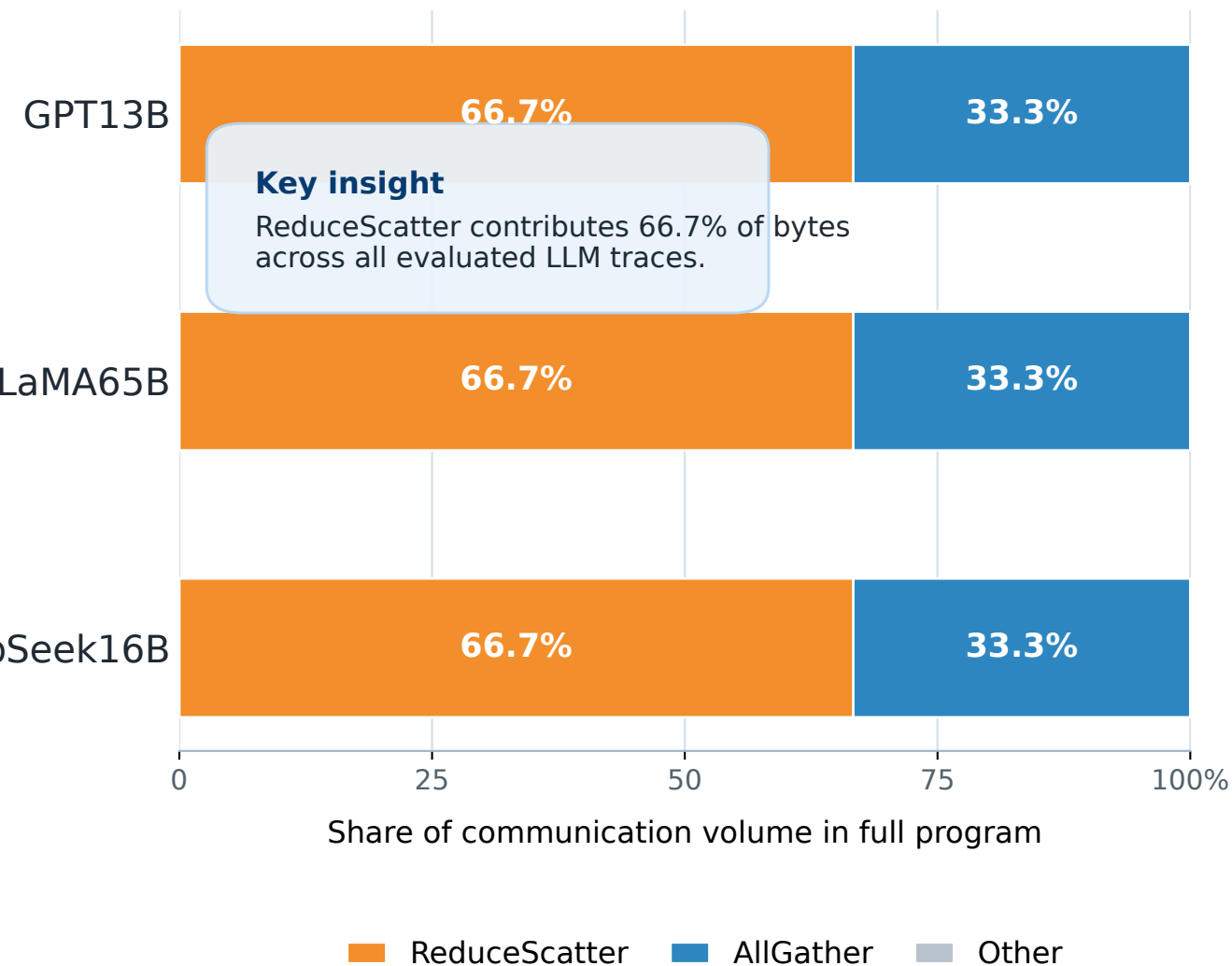


Dominant Collective as a Rank-Mapping Proxy

Optimize only the dominant collective, then evaluate the selected rank mapping on the full communication program.

A. Communication volume concentration



B. Dominant-only matches full-program optimization

# Tenants	Same mapping	Avg JCT Δ	Makespan Δ
2	Yes	+0.0%	+0.0%
3	Yes	+0.0%	+0.0%
4	Yes	+0.0%	+0.0%

$\Delta = (\text{dominant-only result} - \text{full-program result}) / \text{full-program result}$.

0.0%
Avg JCT difference

0.0%
Makespan difference

Scope: evaluated DP traces and tenant counts; the full communication program